Price Mechanism : Demand, Supply, and Market

Price system in a market economy provides an automatic mechanism to allocate and distribute scarce outputs (goods and services) and inputs (resources).

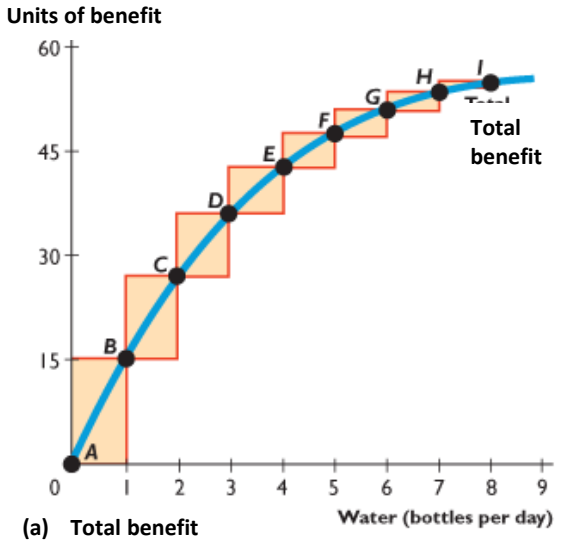
To understand how the price is determined in a market, which consists of all the buyers and sellers of a particular good, we need to consider two forces :

Demand and Supply

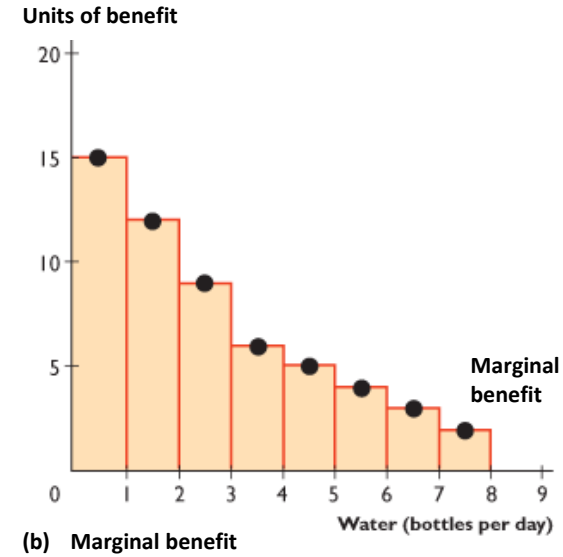
Demand

Demand is the **relationship** between **price** and **quantity demanded** of a commodity (or good).

An individual consumer's (buyer's) demand for a good is determined by the **marginal benefit** he/she can derive from the consumption of each additional unit of the good, which is measured by the **maximum amount he/she is willing to pay for it**.



(a) Total benefit



(b) Marginal benefit

Water (bottles per day)	0	1	2	3	4	5	6	7	8
Total benefit	0	15	27	36	42	47	51	54	56
Marginal benefit		15	12	9	6	5	4	3	2
		A	B	C	D	E	F	G	H	I

Cost-benefit Principle of Demand

Lisa will buy one more cup of coffee if and only if the benefit (marginal benefit) ≥ the cost (price).

If the price is $20 per cup, how many cups of coffee will Lisa buy each day?

2 cups

Quantity	Marginal cost ($)	Total benefit ($)	Marginal benefit ($)
1	20	40	40
2	20	60	20
3	20	75	15
4	20	85	10
5	20	90	5

Cost-benefit Principle of Demand

Lisa will buy one more cup of coffee if and only if the benefit (marginal benefit) ≥ the cost (price).

How will her quantity demanded change when the price of coffee changes? **Price = $15 ⇒ 3 cups**

Quantity	Marginal cost ($)	Total benefit ($)	Marginal benefit ($)
1	15	40	40
2	15	60	20
3	15	75	15
4	15	85	10
5	15	90	5

Cost-benefit Principle of Demand

Lisa will buy one more cup of coffee if and only if the benefit (marginal benefit) ≥ the cost (price).

If the price is $10 per cup, how many cups of coffee will Lisa buy each day?

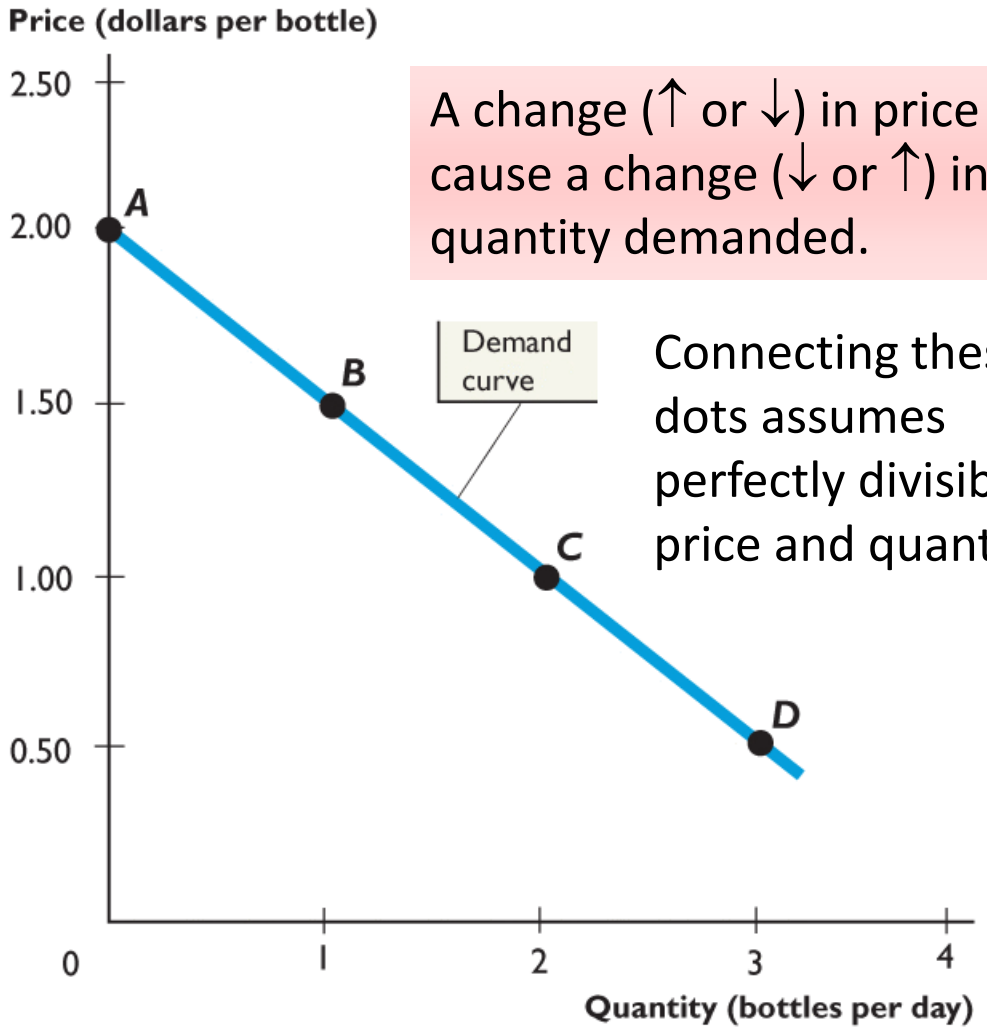
Price = $10 ⇒ 4 cups

Quantity	Marginal cost ($)	Total benefit ($)	Marginal benefit ($)
1	10	40	40
2	10	60	20
3	10	75	15
4	10	85	10
5	10	90	5

Diminishing marginal benefit ⇒ downward-sloping demand curve

Law of Demand: The lower the price of a commodity, the larger the quantity that will be demanded, *other things being equal*, and vice versa.

	Price (dollars per bottle)	Quantity demanded (bottles per day)
A	2.00	0
B	1.50	1
C	1.00	2
D	0.50	3



A change (↑ or ↓) in price will cause a change (↓ or ↑) in quantity demanded.

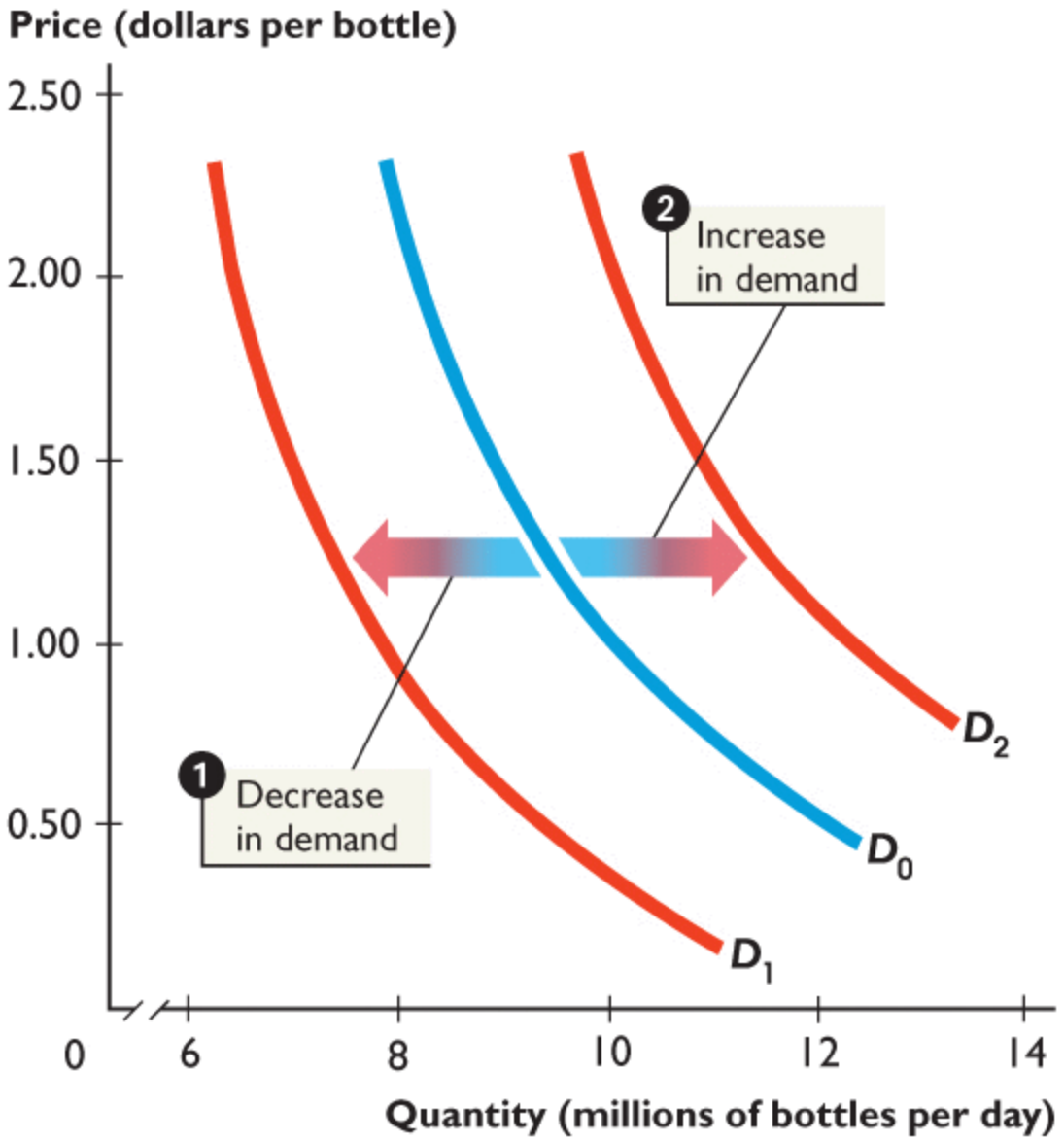
Connecting these dots assumes perfectly divisible price and quantity

A Change in Demand (i.e. a shift in the demand curve)

A change in **individual demand** can be caused by any factor that affects the marginal benefit:

- consumer's income
- price of related goods
- consumer's taste
- expected future price

A change in **market demand** can be caused by any of the above factors and the market size (number of buyers).



A Change in Demand (i.e. a shift in the demand curve)

Reasons for a change in *individual* demand (demand curve to shift):

1. **Income of consumers**

a. Normal good:

An increase in consumer's income causes demand to increase.

b. Inferior good:

An increase in consumer's income causes demand to decrease. They can afford better substitutes in consumption now.

A Change in Demand (i.e. a shift in the demand curve)

2. Prices of related goods

a. *Substitutes in consumption*: e.g., MTR service and Bus service : increase in MTR fare → decrease in *quantity demanded* of MTR service → increase in demand for bus services

b. *Complements in consumption* : e.g., car and petroleum : increase in the price of car → decrease in *quantity demanded* of car ⇒ decrease in demand for petroleum

A Change in Demand (i.e. a shift in the demand curve)

3. Tastes (preference) of consumers

An increased preference by consumers causes demand to increase. E.g., increase in demand for trendy products

4. Expectation of future price

If people expect the price to increase in the future, then they will increase the demand in the current period.

Relevant mainly to investment goods: real property, shares, foreign currencies, gold, etc.

Reason for a change in the *market* demand

A change in the number of consumers/buyers in the market plus the above four factors

A Change in Quantity Demanded vs A Change in Demand

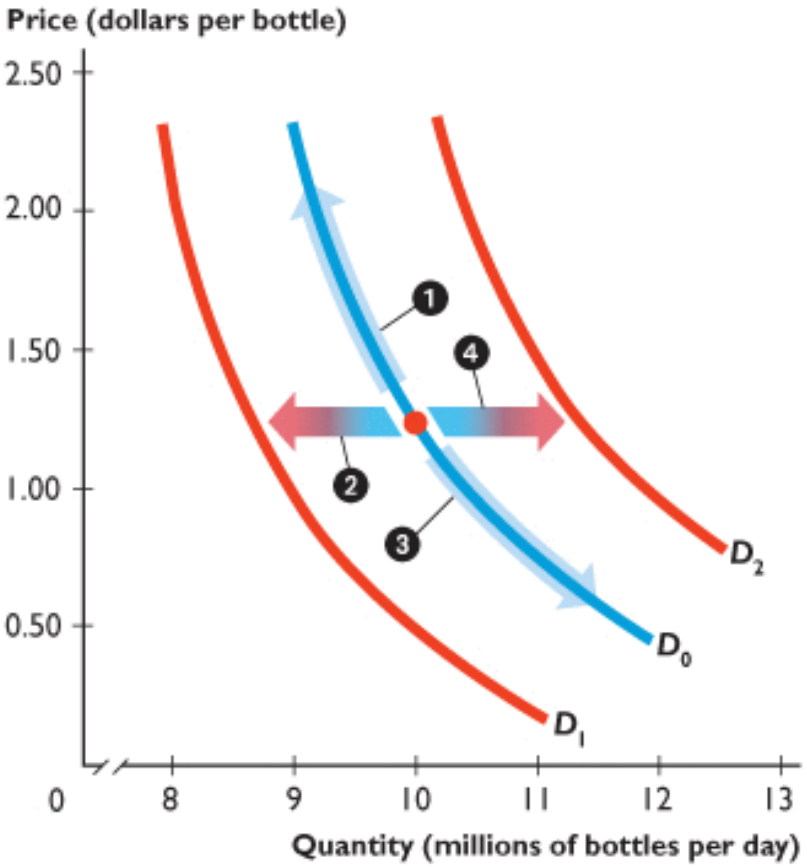
❶ A decrease in the quantity demanded

If the price of a good rises, *cet. par.*, the quantity demanded decreases. There is a movement up along the demand curve D_0.

❷ A decrease in demand

Demand decreases and the demand curve shifts leftward (from D_0 to D_1) if

- The price of a substitute falls or the price of a complement rises.
- The price of the good is expected to fall.
- Income decreases.*
- Expected future income or credit decreases.
- The number of buyers decreases.



❸ An increase in the quantity demanded

If the price of a good falls, *cet. par.*, the quantity demanded increases. There is a movement down along the demand curve D_0.

❹ An increase in demand

Demand increases and the demand curve shifts rightward (from D_0 to D_2) if

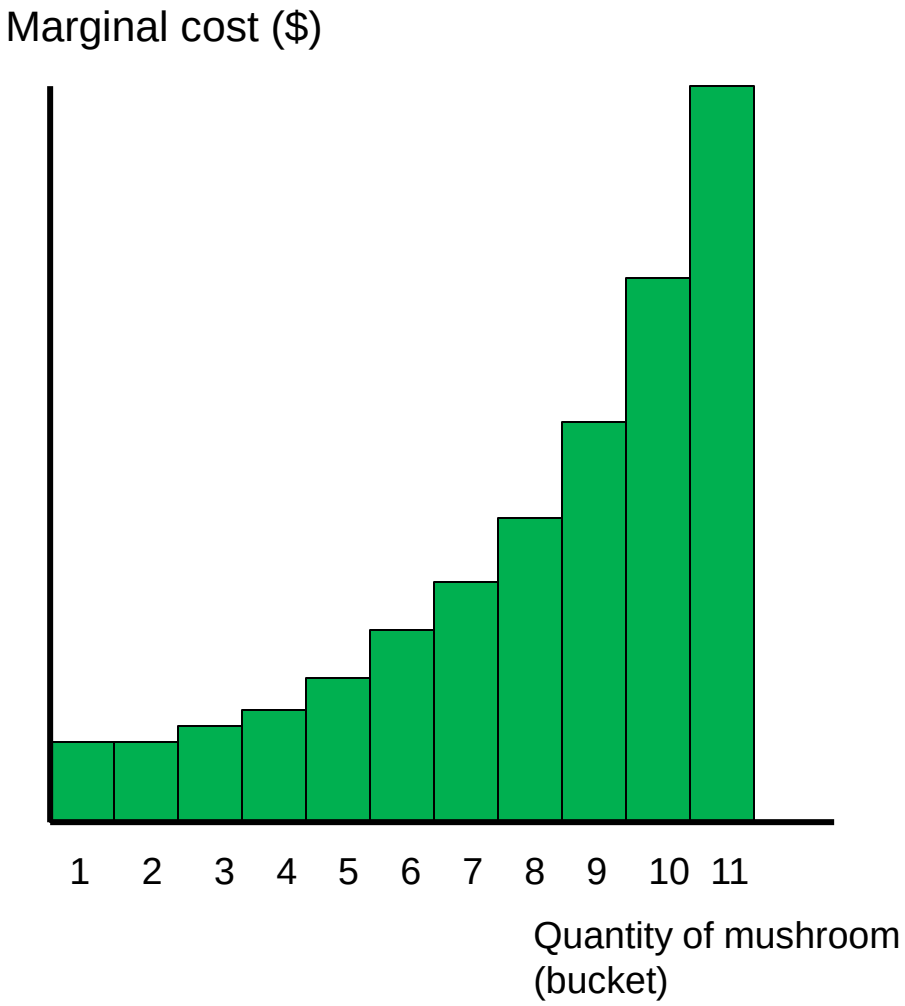
- The price of a substitute rises or the price of a complement falls.
- The price of the good is expected to rise.
- Income increases.
- Expected future income or credit increases.
- The number of buyers increases.

* Bottled water is a normal good.

Supply

Supply is the **relationship** between **price** and **quantity supplied** of a commodity (or good).

An individual producer's or seller's supply of a good is determined by the **marginal cost (MC)** incurred in the production of each additional unit of the good, which determines the **minimum amount the producer or firm must receive** to be willing to produce it.



The farmer will supply one more bucket of mushroom if and only if the benefit (price) ≥ the cost (MC).

Cost-benefit Principle of Supply

You will supply one more hour of tuition service if and only if the benefit (price) ≥ the cost (MC).

If the price is $100 per hour, how many hours of tuition service will you provide each week? **2 hours per week**

Quantity (hours per week)	Marginal benefit ($)	Total cost ($)	Marginal cost ($)
0	-	0	-
1	100	30	30
2	100	100	70
3	100	220	120
4	100	400	180
5	100	650	250

Cost-benefit Principle of Supply

You will supply one more hour of tuition service if and only if the benefit (price) ≥ the cost (MC).

How will your quantity supplied change if the price of tuition service changes? **Price = $150 ⇒ 3 hours per week**

Quantity (hours per week)	Marginal benefit ($)	Total cost ($)	Marginal cost ($)
0	-	0	-
1	150	30	30
2	150	100	70
3	150	220	120
4	150	400	180
5	150	650	250

Cost-benefit Principle of Supply

You will supply one more hour of tuition service if and only if the benefit (price) ≥ the cost (MC).

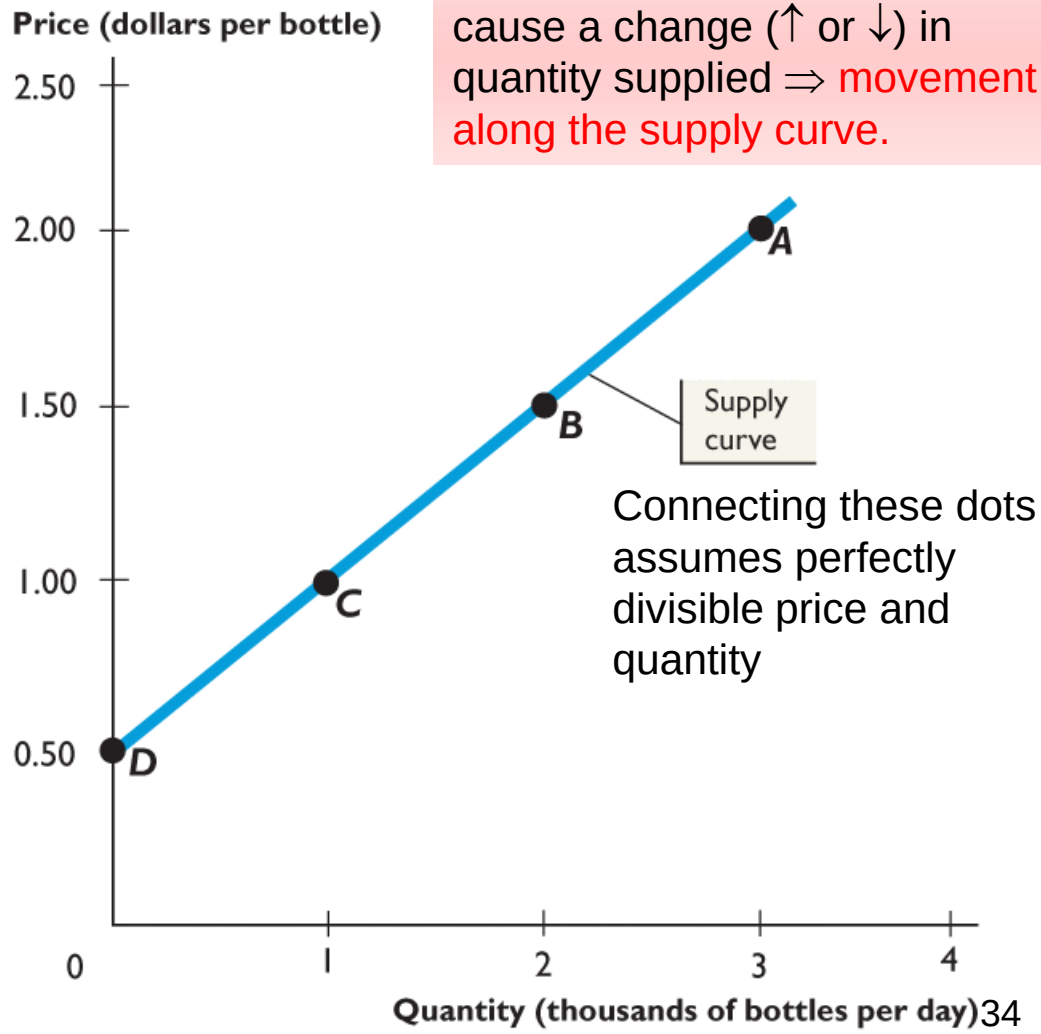
How will your quantity supplied change if the price of tuition service changes? **Price = $200 ⇒ 4 hours per week**

Quantity (hours per week)	Marginal benefit ($)	Total cost ($)	Marginal cost ($)
0	-	0	-
1	200	30	30
2	200	100	70
3	200	220	120
4	200	400	180
5	200	650	250

Increasing marginal cost ⇒ upward-sloping supply curve

Law of Supply: The lower the price of a commodity, the smaller the quantity that will be supplied, *other things being equal*, and vice versa.

	Price (dollars per bottle)	Quantity supplied (thousands of bottles per day)
A	2.00	3
B	1.50	2
C	1.00	1
D	0.50	0



A change (↑ or ↓) in price will cause a change (↑ or ↓) in quantity supplied ⇒ movement along the supply curve.

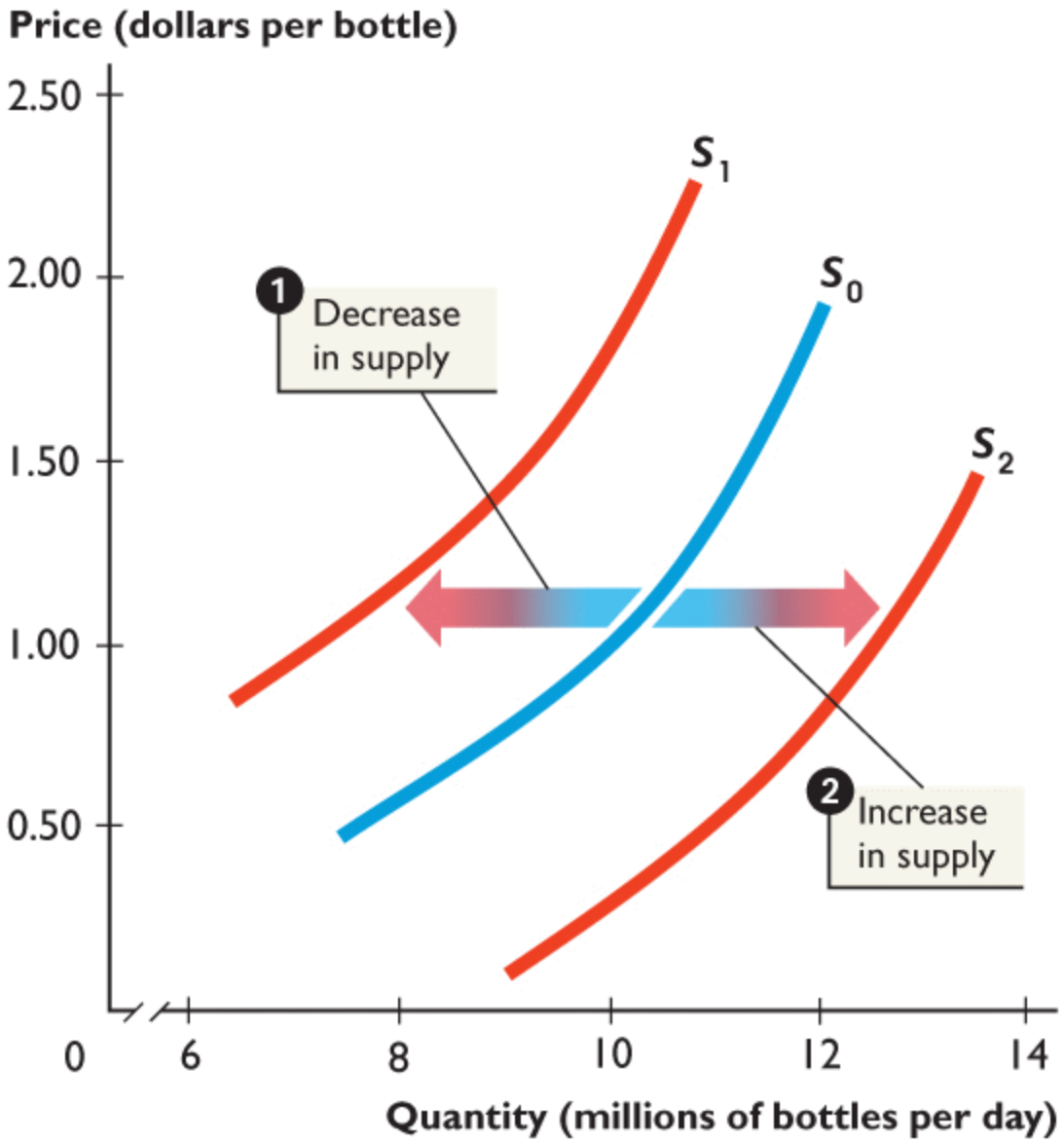
Connecting these dots assumes perfectly divisible price and quantity

A Change in Supply (i.e. a shift in the supply curve)

A change in **individual supply** can be caused by any factor that affects the marginal cost:

- cost of production
- technology (productivity)
- price of related goods
- expected future price

A change in **market supply** can be caused by any of the above factors and the market size (number of sellers).



A Change in Supply (i.e. a shift in the supply curve)

Reasons for a change in *individual* supply (supply curve to shift):

1. **Costs of production** wages, rent, energy prices, etc.

   If production cost increases, then suppliers would ask for a higher price per unit of goods sold, this means supply curve would shift to the left (i.e. decrease in supply).

2. **Technology**

   Technology improvement would reduce the cost of production and cause the supply curve to shift to the right (i.e. increase in supply).

A Change in Supply (i.e. a shift in the supply curve)

3. Price of related goods

a) Complements in production: e.g. beef and leather

Increase in price of beef → ***quantity supplied*** of beef will increase → ***increase in supply*** of leather

b) Substitutes in production: e.g. lettuce and broccoli

Increase in price of lettuce → ***quantity supplied*** of lettuce will increase → ***decrease in supply*** of broccoli

4. Expected future price

If suppliers expect future price to drop, then they will increase the supply in the current period, such as property.

Reason for a change in the *market* supply

A change in the number of suppliers in the market plus the above four factors

A Change in Quantity Supplied vs A Change in Supply

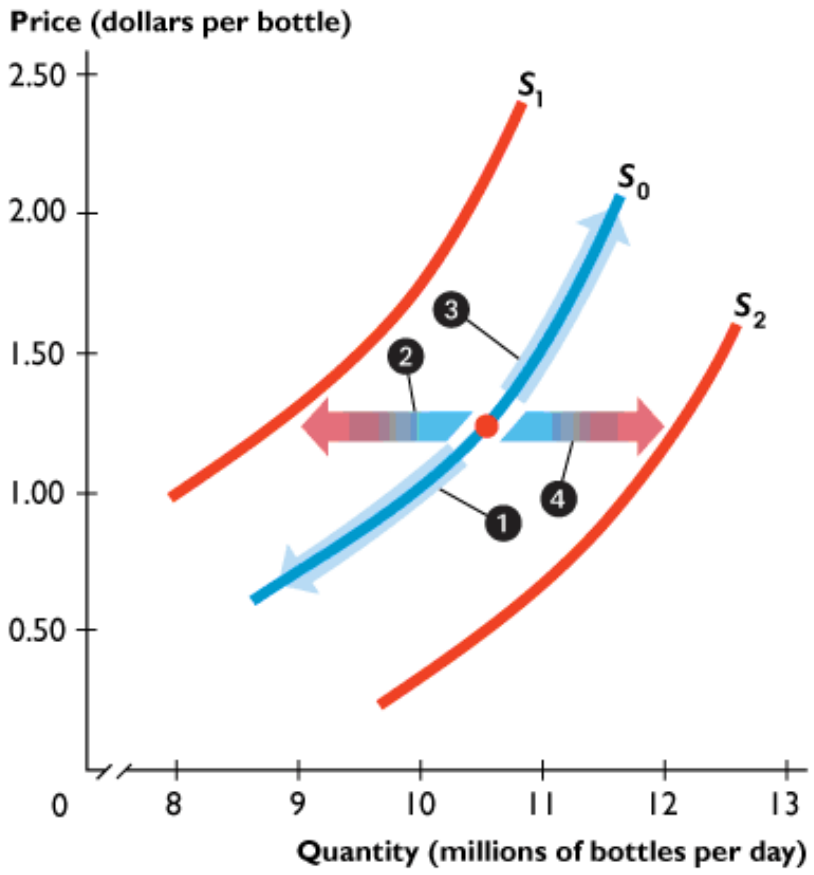
1 A decrease in the quantity supplied

If the price of a good falls, *cet. par.*, the quantity supplied decreases. There is a movement down along the supply curve S_0.

2 A decrease in supply

Supply decreases and the supply curve shifts leftward (from S_0 to S_1) if

- The price of a substitute in production rises.
- The price of a complement in production falls.
- A resource price or other input price rises.
- The price of the good is expected to rise.
- The number of sellers decreases.
- Productivity decreases.



3 An increase in the quantity supplied

If the price of a good rises, *cet. par.*, the quantity supplied increases. There is a movement up along the supply curve S_0.

4 An increase in supply

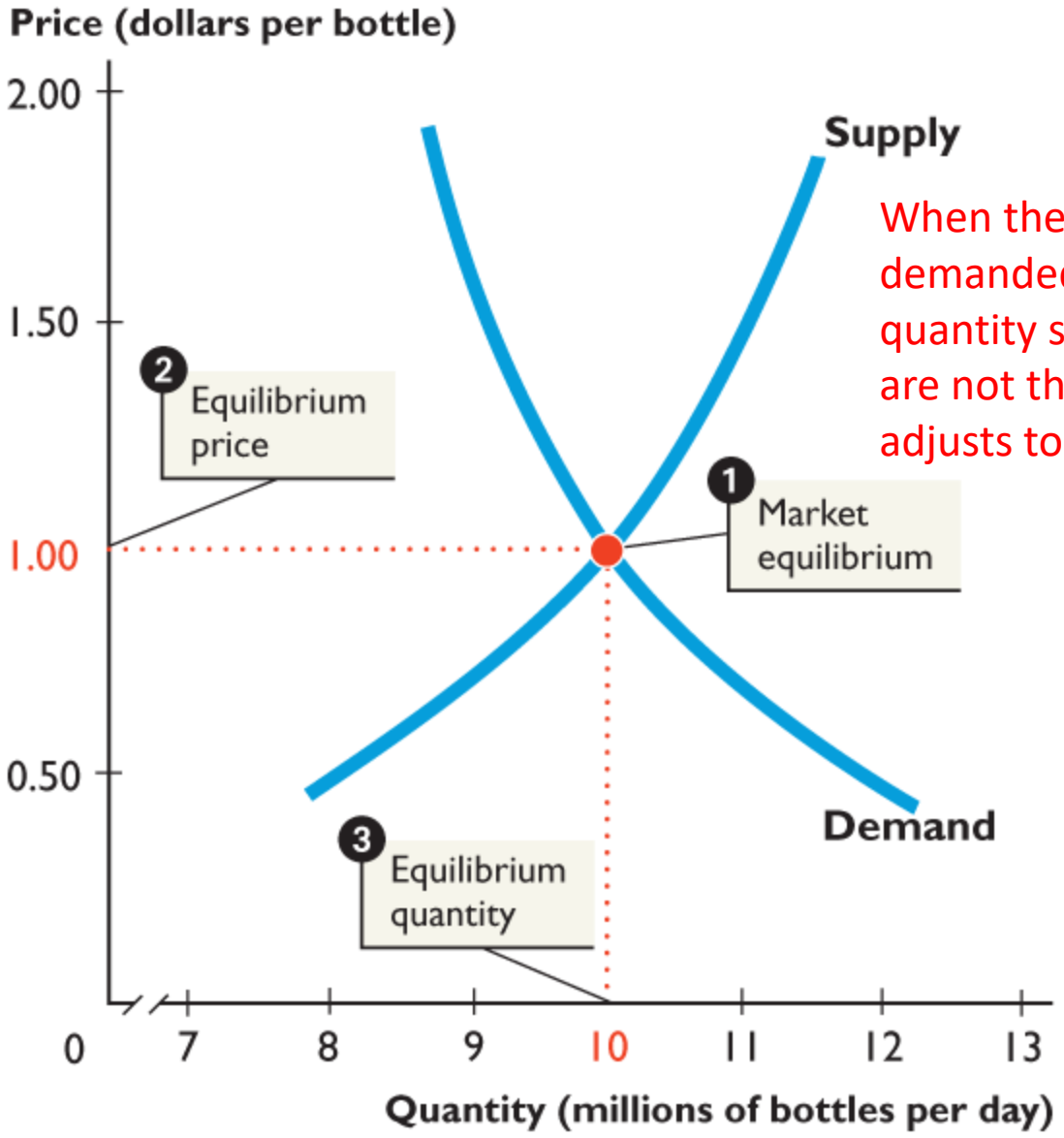
Supply increases and the supply curve shifts rightward (from S_0 to S_2) if

- The price of a substitute in production falls.
- The price of a complement in production rises.
- A resource price or other input price falls.
- The price of the good is expected to fall.
- The number of sellers increases.
- Productivity increases.

Self-assessment of your understanding:

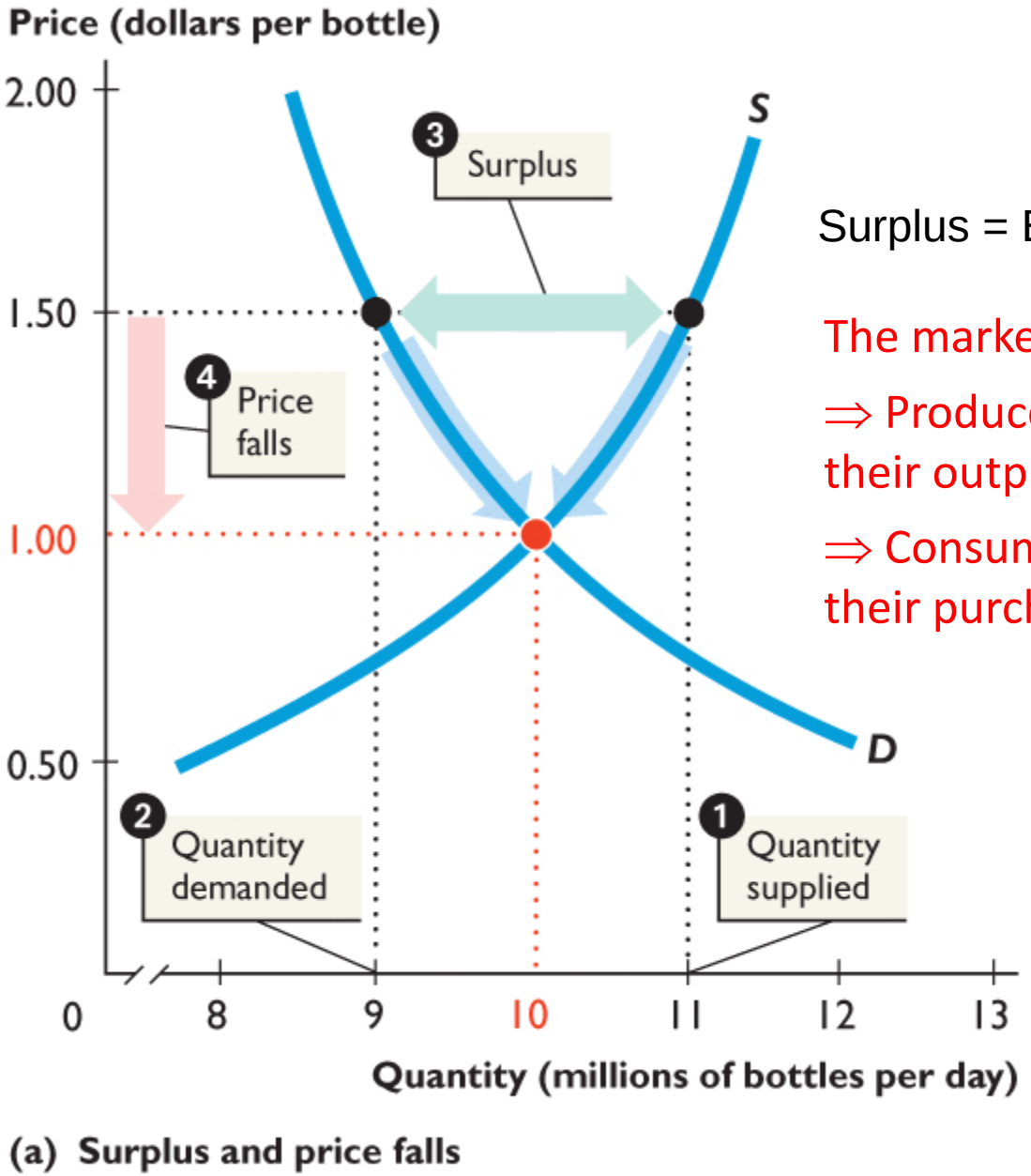
Do Self Exercises 5-7 in Self-study Notes Unit 1.

Market price is determined by the interaction of Demand and Supply.



When the quantity demanded (Q_d) and the quantity supplied (Q_s) are not the same, price adjusts to equalize them.

Automatic adjustment to equilibrium



(a) Surplus and price falls

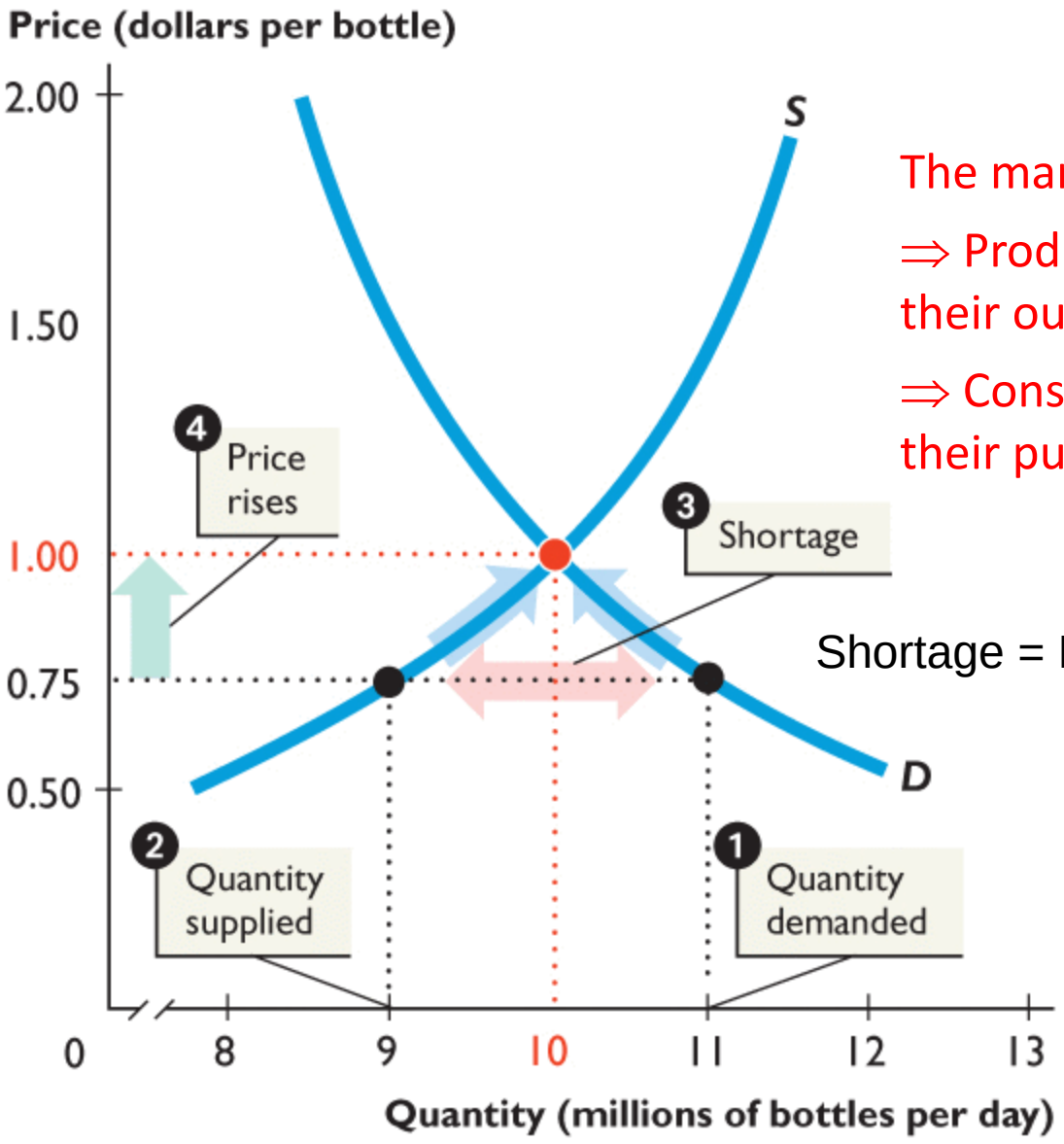
Surplus = Excess supply

The market price drops

$\Rightarrow$ Producers reduce their output ($Q_s \downarrow$)

$\Rightarrow$ Consumers increase their purchase ($Q_d \uparrow$)

Automatic adjustment to equilibrium



(b) Shortage and price rises

The market price rises

⇒ Producers increase their output (Q_s↑)

⇒ Consumers reduce their purchase (Q_d↓)

Shortage = Excess demand

Motivations

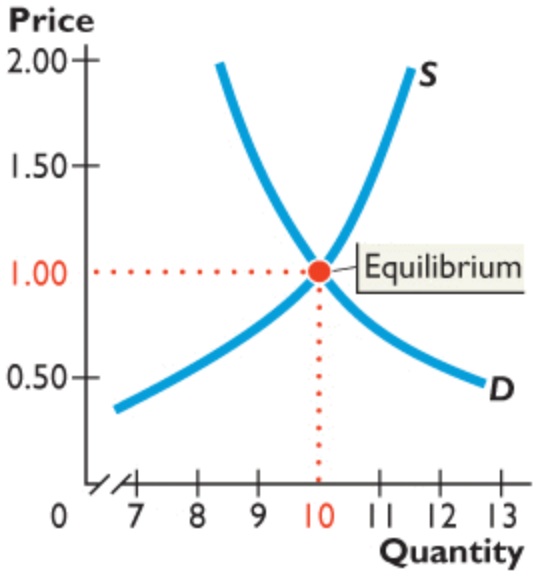
- Why would the share price of CCB drop when there was an IPO of ABC?



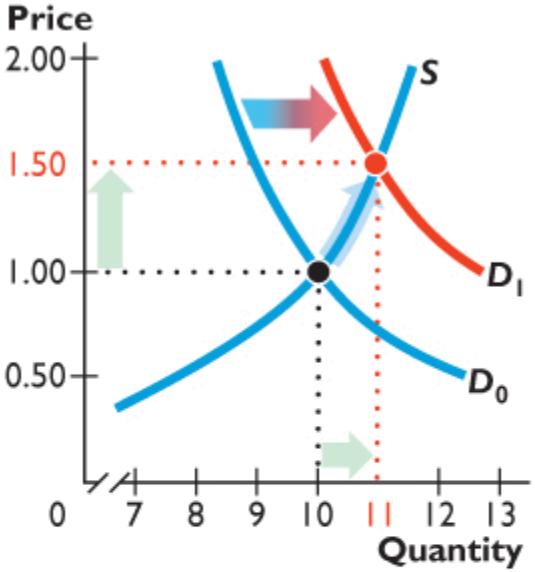


Changes in Equilibrium

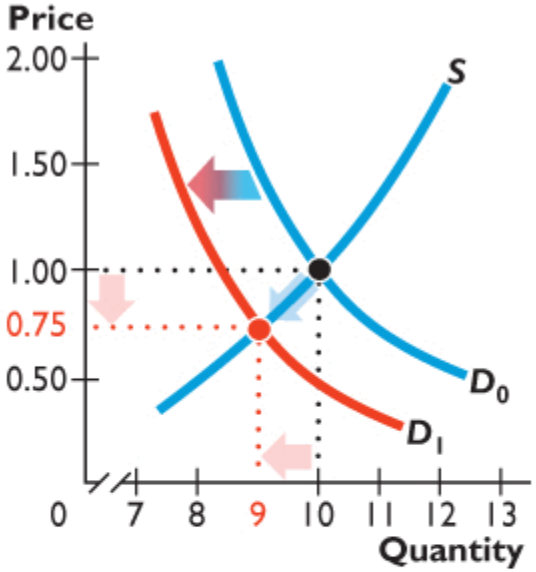
When only demand changes but supply remains the same, the equilibrium P and Q will change in the same direction.



(a) No change in demand or supply



(b) Increase in demand

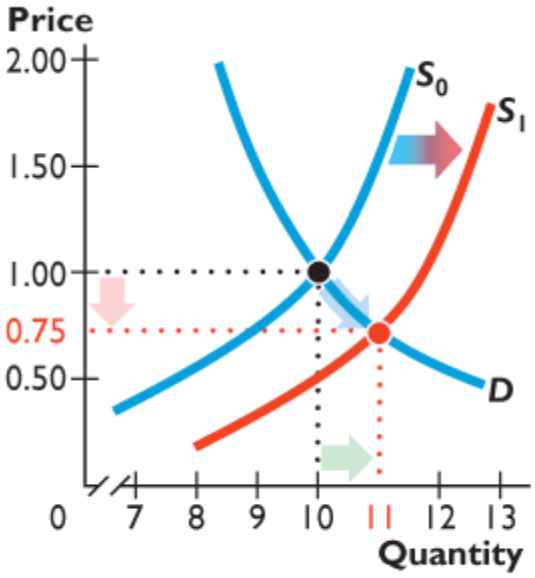


(c) Decrease in demand

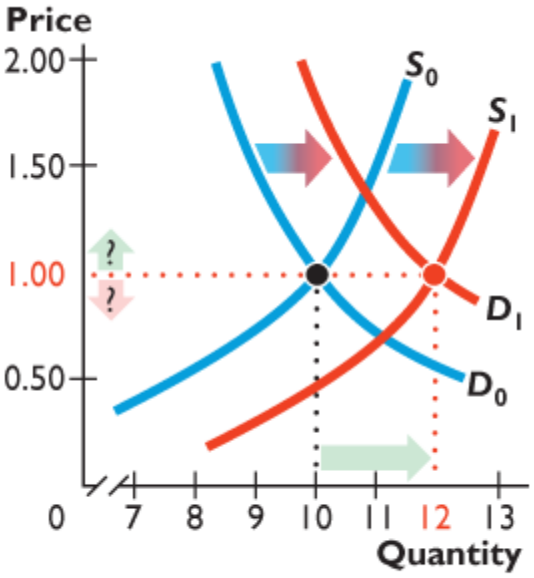
Changes in Equilibrium

When only supply changes but demand remains the same, the equilibrium P and Q will change in opposite direction.

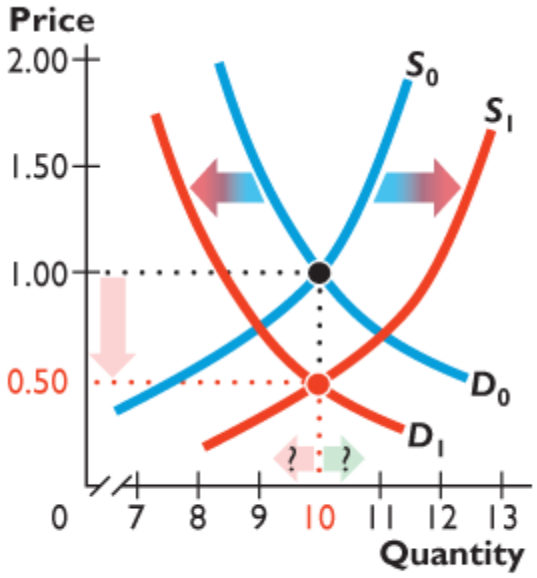
When both demand and supply change, the direction of the change in either P or Q is uncertain, depending on whether D or S changes more.



(d) Increase in supply



(e) Increase in both demand and supply



(f) Decrease in demand and increase in supply

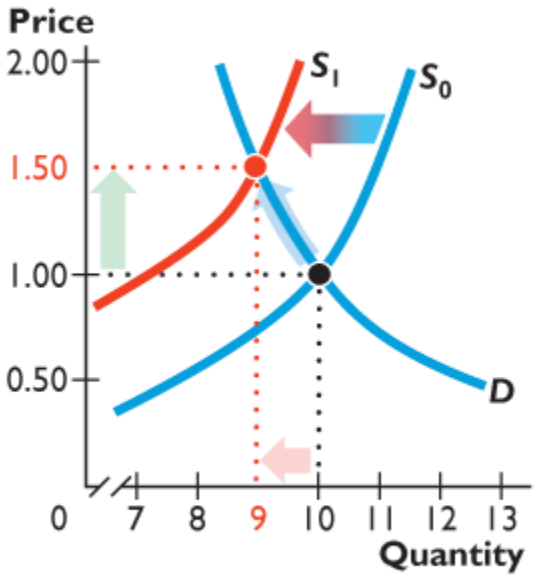
$D\uparrow \Rightarrow P\uparrow \& Q\uparrow$
$S\uparrow \Rightarrow P\downarrow \& Q\uparrow$

$S\uparrow \Rightarrow P\downarrow \& Q\uparrow$
$D\downarrow \Rightarrow P\downarrow \& Q\downarrow$

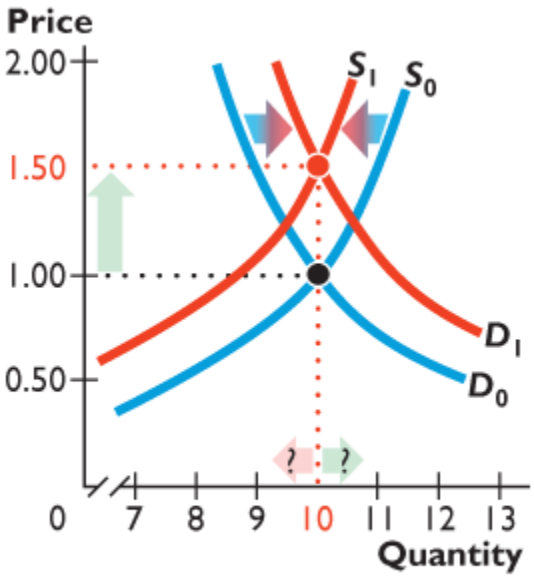
Changes in Equilibrium

When only supply changes but demand remains the same, the equilibrium P and Q will change in opposite direction.

When both demand and supply change, the direction of the change in either P or Q is uncertain, depending on whether D or S changes more.



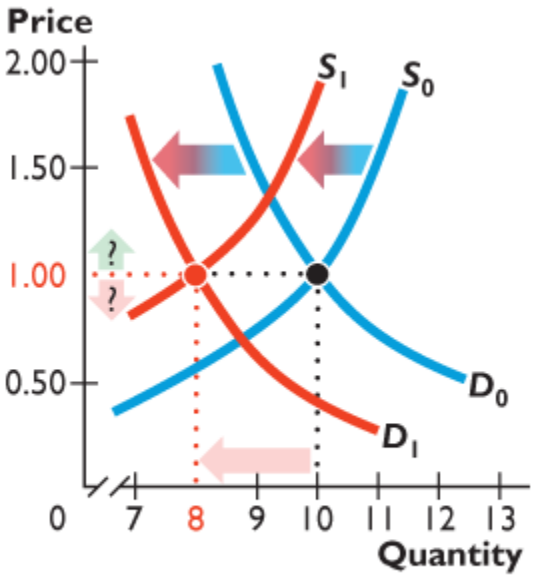
(g) Decrease in supply



(h) Increase in demand and decrease in supply

$D\uparrow \Rightarrow P\uparrow$ & $Q\uparrow$

$S\downarrow \Rightarrow P\uparrow$ & $Q\downarrow$



(i) Decrease in both demand and supply

$D\downarrow \Rightarrow P\downarrow$ & $Q\downarrow$

$S\downarrow \Rightarrow P\uparrow$ & $Q\downarrow$

Which of these diagrams describes what happened to the share price of CCB in the question?